



Met Office



Met Office Careers:

Data Science and Meteorological Phenomenon

Project Brief



Recap

You've heard from Damian, but just in case you wanted a recap of the task, let's take a look at exactly what you need to do.

Sprint Task:

Review the data showcasing temperature differences between two places - St James's Park, in the middle of London, and Northolt, an airfield on the fringe of the city.

Following this, you need to answer two key questions:

- What are the basic relationships within the data?
- What do these relationships mean?

[!\[\]\(3211b5d1d968fc1665909b34f9f16010_img.jpg\) Data Spreadsheet](#)

Things to know

To help complete the task, let's examine some additional information or key terms that are useful to know.

Meteorological Observations

Meteorology is the science that deals with the atmosphere and its phenomena, such as weather and climate. Meteorological observations are crucial for understanding and predicting the weather. Here's how they work:

Weather Stations: These are set up all over the country (and the world!) and equipped with instruments like thermometers for temperature, barometers for air pressure, anemometers for wind speed, and hygrometers for humidity. These stations collect data continuously and are usually situated between 1.5 m and 2 m above the ground surface. Since atmospheric pressure varies with the altitude of the observing location, we correct this value to what it would be if it were measured at sea level.

Satellites: Orbiting the Earth, satellites observe large areas of the world in a variety of different wavelengths, which can be used to infer atmospheric conditions from space. They provide a global view and track storm systems, cloud cover, and more.

Radars: Used primarily to detect precipitation, radars send out radio waves and measure the return signal bouncing off raindrops, snowflakes, or hailstones. This helps meteorologists see where rain is falling and how intense it is.

Cloud lidars: These instruments are a bit like radars but use light waves instead of radio waves and point straight upwards. They record the reflection of the light as it scatters back towards the ground from the cloud, and as the cloud drifts over the lidar, the fractional cover of the cloud and its height can be measured. We report cloud fraction in "oktas", that is "eighths", so a cloud fraction of eight oktas means that the sky is fully covered in cloud and a cloud fraction of four oktas means it is half covered. Sometimes, you'll see a "9" in a report, which usually means there is fog at the surface.

Weather Balloons: These are launched into the atmosphere carrying instruments that send back data on temperature, humidity, and pressure as they rise. From their drift, we can also infer the wind speed and direction. The atmosphere changes with height, and this observing system provides a vertical profile of the atmosphere.

Supercomputers and Models: All the collected data is fed into supercomputers that first provide a complete picture of the weather around the world that is happening now, and then from this state, run physical models of the atmosphere to predict how these conditions evolve in a short space of time. By stepping forward this prediction many times, we can forecast the weather for hours to days ahead. Climate projection works in a similar way, stepping forward from an initial condition but over a much longer period of

time and representing many more physical processes and feedbacks that occur in the climate system, as well as needing to use assumptions about human-related changes such as emissions and land-use.

If you want to put any notes down, you can write them here!

SPRINGPOD — MET OFFICE DATA SCIENCE

Sprint Task:

Analyze temperature differences between St James's Park and Northolt.

Questions:

1. What are the basic relationships within the data?
2. What do these relationships mean?

Key concepts:

- Weather stations collect atmospheric measurements.
- Satellites provide large-scale observations.
- Radar detects precipitation.
- Cloud lidar measures cloud fraction and height.
- Weather balloons provide vertical atmospheric profiles.
- Supercomputers and models use observations to produce weather forecasts and climate projections.



Urban Heat Islands (UHIs)

Urban Heat Islands describe situations where a city is significantly warmer than its rural surroundings. This happens due to human activities and the way cities are built. Here's why UHIs form:

- **Concrete and Asphalt:** Buildings, roads, and other infrastructure absorb and retain heat more than natural landscapes like forests or fields. These materials release their heat slowly too.
- **Reduced Vegetation:** Trees and plants cool the air through a process called transpiration, through the latent heat as water is released into the atmosphere as vapor. Urban areas often lack green spaces, leading to less cooling.
- **Energy Consumption:** Air conditioners, cars, factories, and other sources generate heat. The energy used to cool buildings also contributes to the warmth outside.
- **Building Density:** Tall buildings and narrow streets change the way in which air flows through the city and the way that air from aloft can be mixed with air at the ground surface.



Impacts of UHIs:

- **Health:** Higher temperatures can lead to heat-related illnesses, such as heat stroke or dehydration, especially during heatwaves.
- **Energy Use:** More energy is needed for cooling, increasing electricity demand and potentially leading to power outages.

- **Air Quality:** Reduced ventilation at ground level and higher city emissions can worsen air pollution, creating more smog and affecting respiratory health.

Mitigating UHIs:

- **Green Roofs and Walls:** Planting vegetation on buildings helps cool the air and provides insulation.
- **Urban Planning:** Designing cities with more parks, green spaces, and trees can significantly reduce UHI effects.
- **Reflective Materials:** Using materials that reflect rather than absorb heat for buildings and roads can lower temperatures.
- **Cool Roofs:** Roofs designed to reflect more sunlight and absorb less heat can help keep buildings and cities cooler.

Understanding meteorological observations helps us predict weather and climate changes, while awareness of UHIs can lead to better urban planning and healthier, more sustainable cities.



Project Activity: Working with the Data

It's important to understand that there are different ways to work with this data. The following represents a simplified workflow that a data scientist would use when exploring relationships between data.

- **Do we understand what the data is about?**

This sounds like a simple question, but at its core, this is about being able to understand what is being reported in each column of the Data tab in the spreadsheet.

For support using Excel commands, please refer to the tip sheet linked below.

[!\[\]\(83f22ed94ec5517769dd76d702c6bfd8_img.jpg\) Using Excel Commands](#)

Here are some questions we need you to consider. You can answer them in the boxes below.

How is the temperature observed?

The temperature is observed using a thermometer at the weather station and recorded in degrees Celsius (°C).

What does cloud amount mean?

cloud amount means how much cloud is covered in sky measured in oktas where 0 oktas means clear sky and 8 means cluster of clouds and 9 mean fog at surface.

What happens to the data when we change the clocks from Greenwich Mean Time (GMT) to British Summer Time (BST)?

When the clocks change from GMT to BST, the recorded times move forward by one hour. The actual weather observations do not change; only the time shown for the observations changes.

Is the data complete? Does it contain any unusual values (such as missing values, or not applicable)?

The data is not completely complete because NA values are present in the dataset. These indicate that some observations are not available. There do not appear to be blank cells, but NA is used where a value is unavailable.

If the data were incomplete, what would this mean? What should we do with such entries?

Incomplete data means that some observations are missing or unavailable. We should first investigate why the values are missing and then handle them appropriately. For numerical measurements such as temperature, missing values could be replaced with the median if appropriate, as it is less affected by unusual values than the mean.

What might the Met Office do to check that the data being recorded is correct?

The Met Office could check the data using quality-control checks, such as looking for missing values, unusual or unrealistic readings, and sudden changes that may indicate a problem with the measurement or equipment.

What does Excel do when it encounters such an entry?

Excel treats an NA entry as text rather than as a numerical value. Therefore, functions such as AVERAGE do not treat it as a temperature measurement.

- **What are the basic relationships within the data?**

Once we have understood what the data is showing, check out the Hour and Month tab of the spreadsheet.

Here, the difference in temperature between St James's Park and Northolt has been averaged according to the month of the year (columns B to M), and the hour of the day (rows 7 to 30). So, midnight is 0, and noon is 12. For instance, if we were to look at cell G12 – this is the average difference in temperature for 0500 during June.

Within the sheet, you will find average temperature differences over all months as a function of hour of day in column N, and the average temperature difference over all times as a function of month in row 31. Try and plot these relationships. Take a look at the following questions, and answer them below.

Does the strength of the urban heat island change more with the time of year, or by the time of day?

The strength of the urban heat island changes more with the time of day than with the time of year. The temperature difference varies by about 1.03°C across the hours of the day, compared with about 0.38°C across the months.

Can you plot the change in temperature difference (y-axis) as a function of the time of day (x-axis) for the two different months of January and July? Do they look the same? What might be going on?

No, January and July do not show exactly the same pattern. Both months show that the temperature difference changes throughout the day, but the size and shape of the changes are different. The urban heat island effect is generally stronger at some times of the day than others. This may be related to differences in heating and cooling between the urban area and the surrounding area, which vary with daylight, temperature and other weather conditions.

Similarly, the “wind speed and cloud” tab shows the same data but the average has been worked out according to the wind speed and cloud amount. Variation in wind speed has also been plotted (column K against column A).

Which of the three factors that we have looked at (hour of day, month of year and wind speed) is the most significant in determining when the urban heat island be the strongest?

Wind speed appears to be the most significant of the three factors. The temperature difference varies by about 2.32°C with wind speed, compared with 1.03°C by hour of day and 0.38°C by month of year. The temperature difference generally decreases as wind speed increases.

What might be causing this effect?

The urban heat island is likely stronger when wind speeds are low because there is less movement and mixing of air between the warmer urban area and the surrounding area. As wind speed increases, the air is mixed more effectively, reducing the temperature difference between St James's Park and Northolt.



Project Activity: Extra Challenge

Those of you who are very good at maths might further want to think about the best type of functional form that could be used to fit the urban heat island data, with a view to fitting the data. It's tricky to use Excel to fit the data, data-scientists would usually look to programming languages such as R or Python to do this, although Excel can do some of this using the "Solver" add-in if it's available to you.



Project Debrief

Main Points

- Without data, we would not be able to add any physical context to the weather that we experience.
- If we gather enough good-quality data, we can start to see relationships between different quantities. In this case, it was the urban heat island, but it could be any relationship at all, for example it might be using weather data to predict health problems. But not everything is predictable just from your data sets!
- When you next see a weather forecast being presented, look out for the presenter giving night-time minimum temperatures that are different for cities than for the surrounding countryside. They are talking about the urban heat island, which you've just explored.



Download Your Workbook



You can submit these files to your school or college as evidence of the work that you've completed, and may want to submit them to the Met Office, alongside the virtual work experience workbook, if you want to apply for an in-person work experience opportunity in the future.